

Leatt Ecommerce Intelligence Story

From public ecommerce data to Fabric Lakehouse, Power BI, ML, AI SEO and business decisions

Built	Proof
2,000,000 transaction lines	Fabric / OneLake upload proof
R3.99bn modeled revenue	Power BI semantic model and reports
48.5% gross margin rate	Decision intelligence and KPI rationale
4.9% return leakage	ML return-risk watchlist
Fabric F2 paused after export	Cost-control handover

1. The Business Question

The story starts with a practical executive question: if Leatt had to run a modern ecommerce intelligence program, what data would leadership need to protect growth, margin and customer experience?

The answer is not just a dashboard. The business needs a governed operating rhythm: extract product and transaction signals, model them in Fabric, expose trusted KPIs in Power BI, use ML to identify leakage, and turn the signals into owner-led actions.

The modeled estate is large enough to feel enterprise-grade: 2,000,000 transaction rows, R3.99bn revenue and R1.94bn gross margin.

2. What The Data Says

The strongest commercial signal is concentration. Apparel is the hero category, contributing R1.76bn. That is good news because it gives leadership a clear growth engine; it is also a risk because availability, SEO visibility and return leakage in that category matter disproportionately.

Margins are attractive at 48.5%, but returns absorb 4.9% of modeled revenue. That makes return reduction one of the cleanest business cases in the pack.

Category	Revenue	Gross Margin	Return Rate
Apparel	R1.76bn	R915.7m	6.2%
Body Protection	R637.3m	R291.9m	3.7%
Helmet	R495.2m	R211.2m	2.9%
Footwear	R474.1m	R187.4m	7.1%
Gloves	R336.9m	R182.3m	3.1%

3. What We Learned

The main learning is that ecommerce BI should not stop at reporting revenue. Revenue only becomes useful when it is reconciled with margin, returns, channel efficiency, stock availability and content quality.

The live storefront evidence points to Shopify rather than SAP Commerce/Hybris. That changes the architecture story: Shopify/public catalog signals can sit upstream, while SAP Business One or SAP BW fits downstream as finance, inventory, VAT and reconciliation data.

dbt is optional for this proof. It would be useful later for versioned SQL transformations and CI tests, but Fabric Data Factory, Lakehouse files, notebooks/scripts and Power BI are enough to demonstrate the senior BI workflow.

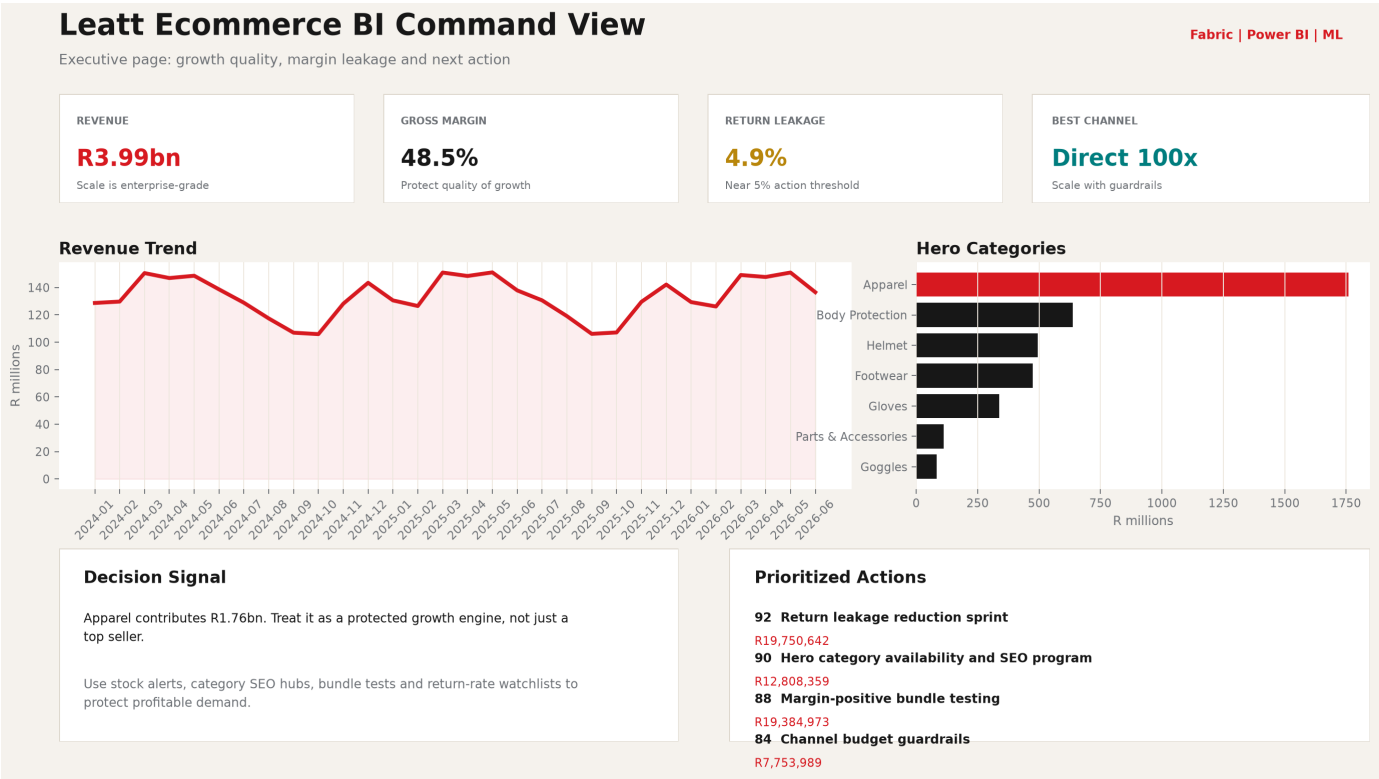
4. What Leadership Should Do Next

The project now turns insights into management actions. Each action has a rationale, owner, value estimate and next step.

Priority	Initiative	Value	Owner	Next step
92	Return leakage reduction sprint	R19,750,642	COO / CX	Create return-reason taxonomy and product-level return dashboard.
90	Hero category availability and SEO program	R12,808,359	CEO / Ecommerce	Build category scorecard with stockout, ranking, margin and landing page signals.
88	Margin-positive bundle testing	R19,384,973	Merchandising / CMO	Run A/B test: product page bundle versus normal recommendation carousel.
84	Channel budget guardrails	R7,753,989	CMO	Set campaign guardrails: minimum ROAS, conversion lift and margin contribution.
82	SAP close reconciliation automation	Risk reduction / faster close	CFO / BI Lead	Schedule monthly exception pack with owner, SLA and SAP account mapping.

5. How The Power BI Report Should Be Read

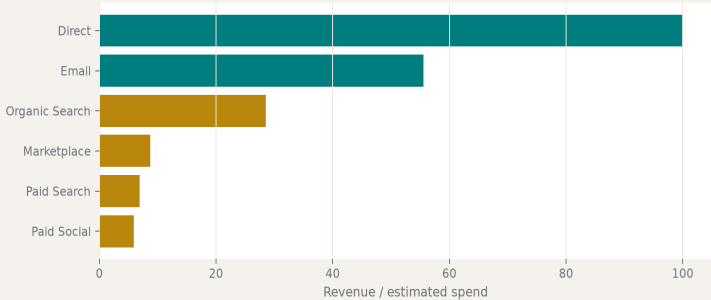
The report is designed as a sequence, not a menu. Start with the executive command view, move to marketing and AI SEO, then finish with ML/governance. This follows the logic of the business: first understand performance, then growth levers, then risk and auditability.



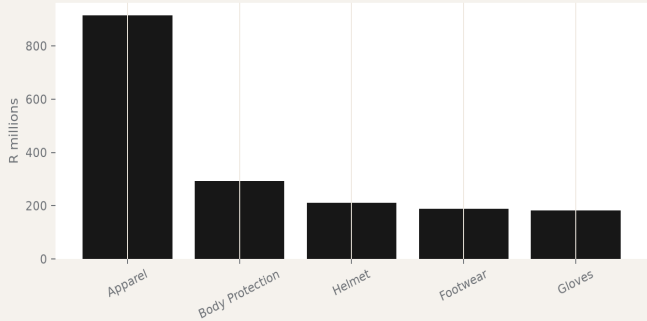
Growth, AI SEO And Channel Quality

Marketing page: spend only scales after ROAS, landing-page relevance and margin hold

ROAS Guardrails



SEO Should Prioritize Margin Pools

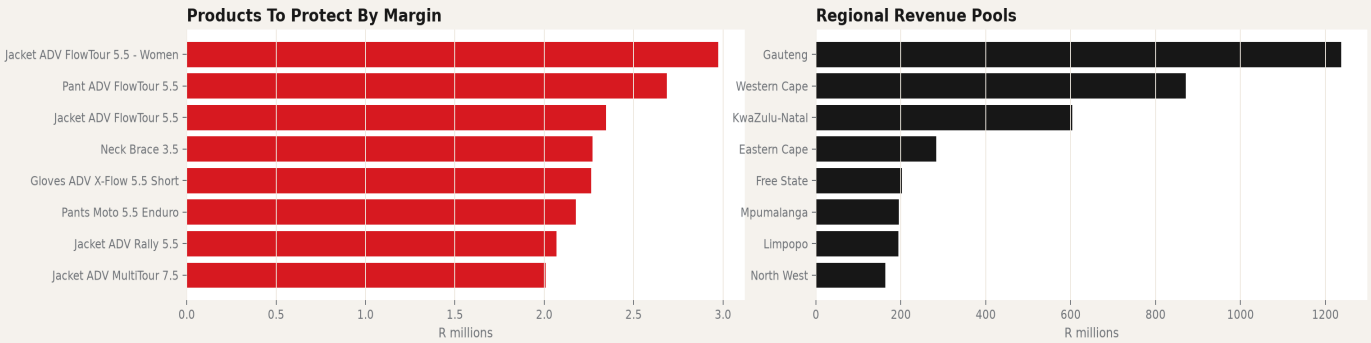


12-Week AI SEO Roadmap



ML, Governance And Audit Readiness

Risk page: return leakage, data controls and finance-ready evidence



Signal Rules: What Changes Trigger Action		
Revenue momentum	R3,994,247,413	Trigger exception commentary in the monthly trading pack.
Margin quality	48.5%	Protect margin-positive product mix and bundle high-margin items.
Return leakage	4.9%	Improve sizing, delivery promise, PDP content and post-purchase guidance.
Hero category concentration	Apparel R1,759,430,045	Create availability alerts and category-specific SEO/story pages.
Channel efficiency	Direct 100.0x ROAS	Move spend toward efficient journeys; keep paid experiments controlled.

6. Signal Rules And Governance

A senior BI solution should say what happens when a metric moves. The signal rules below make the report operational.

Signal	Reading	Rule	Owner	Decision
Revenue momentum	R3,994,247,413	If monthly movement exceeds +/-12%, explain by category, change and exception commentary in the monthly trading pack.	Finance	
Margin quality	48.5%	If margin rate falls below 45%, inspect discounting, COGS and pricing. Create margin-positive product mix and bundle high-margin items.	COGS	
Return leakage	4.9%	If return rate exceeds 5%, prioritize category root cause work to improve sizing, delivery promise, PDP content and post-purchase guidance.	COGS	
Hero category concentration	Apparel R1,759,430,045	If one category exceeds 35% share, manage to stock and campaign. Create category lists and category-specific SEO/story pages.	COGS	
Channel efficiency	Direct 100.0x ROAS	Scale only when ROAS, margin and conversion all stay above thresholds. Move spend toward efficient journeys; keep paid experiments controlled.	COGS	
Data completeness	2,000,000 rows	If row counts differ from expected load by 1%, block semantic publish source, target, row count and commit ID in every close pack.	COGS	

7. AI SEO: The New Growth Layer

The latest search direction is clear: AI search visibility depends on crawlable pages, complete product data, structured data, Merchant Center feed quality, useful content and measurable share of voice. This is not separate from BI; it belongs in the same growth model.

Timeframe	Workstream	Action	Outcome
0-2 weeks	Technical audit	Validate Product/ProductGroup schema, sitemap canonicals, index coverage, server-side rendering, HTML format	Indexable content, correct structured data, Merchant Center feed health.
0-2 weeks	Hero category content sprint	Build an AI-search-ready Apparel hub with buyer questions, use cases, category-specific content and technical links	Non-used category pages, category authority.
2-4 weeks	Product feed enrichment	Improve titles/descriptions, GTIN/SKU coverage, product highlights, shipping, complete policies and shipping visibility.	High quality product data, complete policies and shipping visibility.
2-4 weeks	Facet governance	Define index/noindex/canonical rules for size, color, active price, row and stamp duplication	Reduced crawl and duplicate content risk.
4-6 weeks	AI visibility dashboard	Create Power BI report for AI share of voice, conversational search, product transparency and competitor presence.	Management visibility, product transparency and competitor presence.
4-8 weeks	Profit-based SEO model	Join SEO landing pages to revenue, margin, stock and SEO profit	SEO profit ranked by profit and operational risk.
8-12 weeks	Experiment loop	Run A/B tests for category hub copy, PDP fit guidance, product fit guidance and FAQ AOV test	Measured impact and SEO AOV test return reduction.

8. The Portfolio Message

This project demonstrates the full senior BI developer arc: source research, data engineering, Fabric/F2 capacity use, Lakehouse upload, semantic modeling, Power BI report creation, ML, governance, SAP/accounting thinking, AI SEO and Git proof.

The most important message for an interviewer is this: the work is not just technically broad. It is commercially literate. Every metric has a reason, every report has an audience, and every insight has a next action.

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